062102 - Applied Statistics

Polytechnic University of Milan

A.Y. 2025/26 *class notes*

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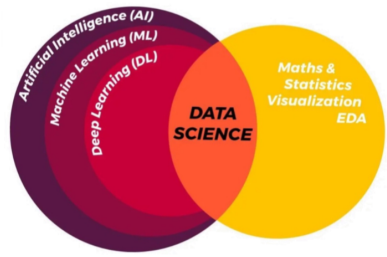
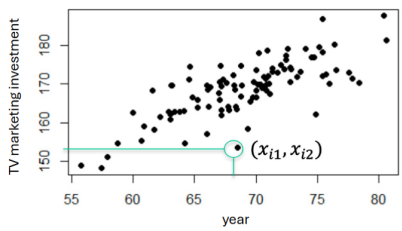
1 Introduction

Statistic is the foundation for all data-driven decision-making processes, also in the AI era: in fact, nowdays machine learning models rely mostly on statistical methods. The main difference resides in the purpouse: *ML models* are designed to make the most accurate prediction possible, *statistical models* instead aim to make inference about the relatioship between the data.



In this course we are going to deal with *multivarieated data*, meaning that each data point contains several variables, defining their *dimensions*. Representing this data with matrices is the more convient way.



As for ML, we have diffent classes of methods based on the available data and how we interpret them:

Supervised learning

$$\mathbb{X} = \begin{bmatrix} x_{1,1} & x_{1,2} & x_{1,n} \\ \dots & \dots & \dots \\ x_{m,1} & \dots & x_{m,n} \end{bmatrix}$$

Y sales — **X** channels

In this class we can idenitify a **target** variable Y and a set of "input" variables X called **features**. Targets and features relationship is expressed trougth a function affected by a certain error ε: $Y = f(X) + \varepsilon$. The error ε is something we can never get rid of it, we should think about it as a property of the data.
Tipically in this set of method we try to make prediction about new target values or finding those features that are more important to explain the targets.

Unsupervised learning

In this case there aren't target varibales, but we all the varibales are features ($f(X) \approx X$).
This set of models focuses on explloting structures and patterns within the data itself.

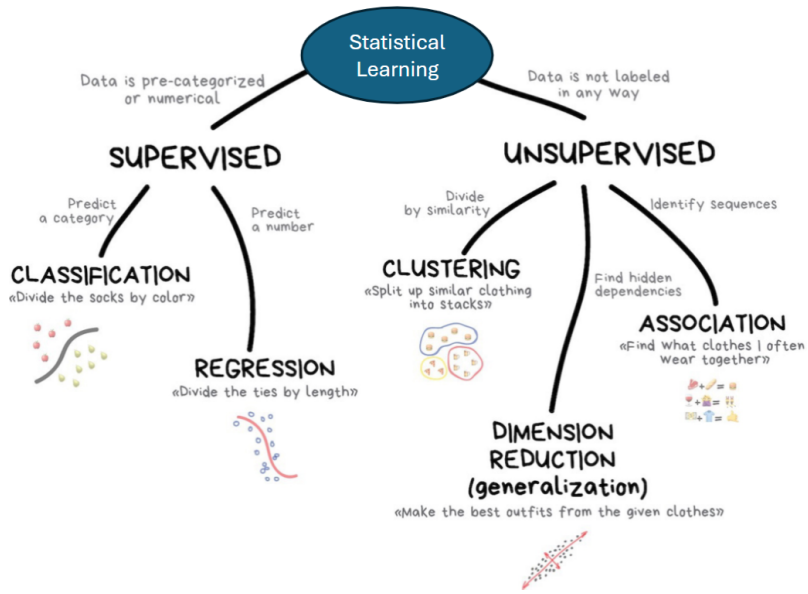


Figure 1: Taxionomy of the methods explored during the course.

1.1 Model goodens

How do we find a good model for our data?
To start, let's take linear models as examples, like $Y = \beta_0 x_0 + \beta_1 x_1 \cdots + \beta_p x_p$.
The models we want to use are expressed trough a set of **parameters** of our choice (in the previous example we have $p + 1$ parameters); the goal is to tune these parameters using training data in order to shape our function accordingly to the data. However most of the times linear models are not complex enough, and we have to shift toward more complex ones.

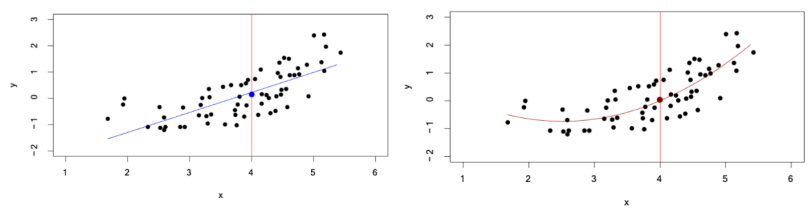


Figure 2: Comparison between linear model ($f(X) = \beta_o + \beta_1 x$) and quadratic model ($f(X) = \beta_0 + \beta_1 x + \beta_2 x^2$).

Usually the choice of a model is influenced by some trade-offs:

- *prediction accuracy vs interpretability* : we might trade-off the model's accuracy to make it more interpretable
- *parsimony principle* : usually we prefer simpler models involving fewer variables since they're easier to manage
- *good fit vs over-fit or under-fit* : see below

1.2 Underfitting vs Overfitting problem

Start with the example of estimating the height of the next student entering the room during lecture, in order to do that our prediction $\hat{y}$ should minimize a certain function, usually the **Mean Squared prediction Error (MSE)**.

$$MSE(\hat{y}) = \frac{1}{n}\sum_{i=1}^{n}(\hat{y}_i - y_y)^2$$

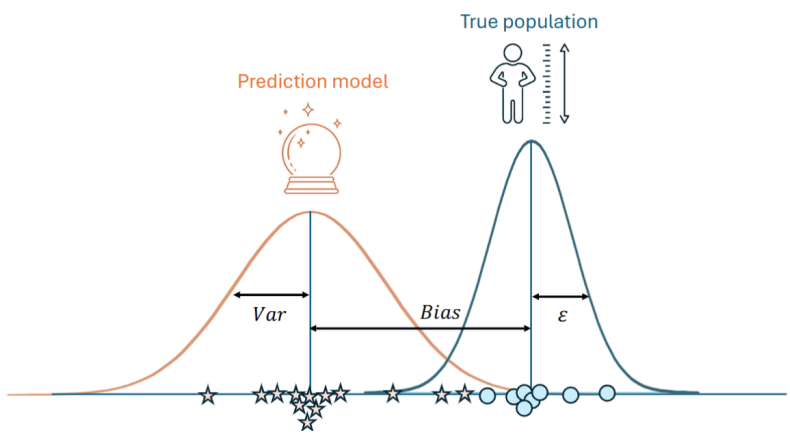
If we know the height of all students, we want to use the population mean μ as predictor, doing so we get as MSE $\mathbb{E}[(Y-\mu)^2] = Var(Y) = \sigma^2$.

The data variance is irriducible, we can't do better than that.

However, it's impossible to have information about the whole population; so what we do is taking a small sample from it over and over, using the sample mean $\bar{Y}$ as μ and getting $MSE_{avg} = \mathbb{E}[(\hat{Y}-Y)^2]$. By decomposing the average MSE we are able to understand which are our sources of error: this reduction leads to two source of errors, *bias* and *variance*.

Bias : $Bias*(\hat{Y}) = \mathbb{E}[\hat{Y}-\mu]$, we have bias if our predictions are systematically to high or too low

Variance : $Var(\hat{Y}) = \mathbb{E}[(\hat{Y}-\mathbb{E}[\hat{Y}])^2]$, we have high variance if our predictions are very sensitve on the sampling we draw from the universe



So if we try to fit a model $f(X)$ with some training data, and we end up with $\hat{f}(X)$ with some test observation (x_0, y_0), we have that

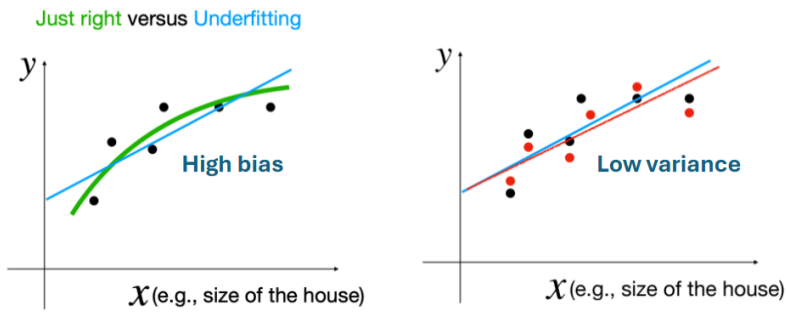
$$\mathbb{E}[y_0 - \hat{f}(x_0)] = Var[\hat{(f)}(x_0)] + Bias[\hat{f}(x_0)]^2 + Var(\varepsilon)$$

Varince : captures how much our model changes based on the training dataset

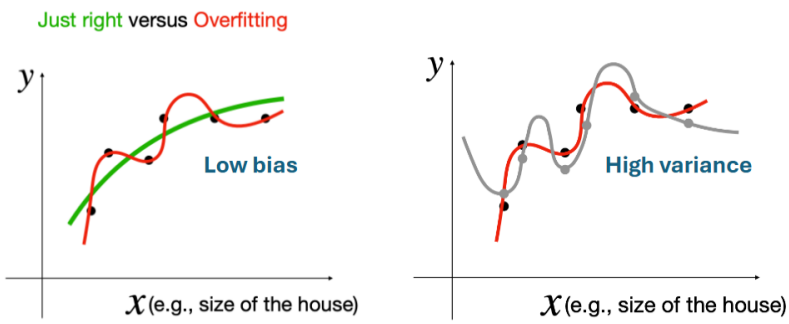
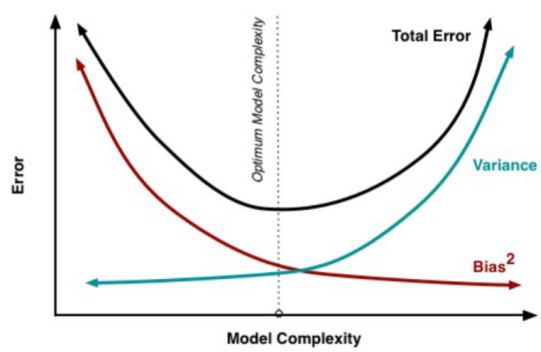
Bias : the inherith error we get from our models

Noise : ambiguity due the data distribution

If we take a model too much simple, we end up with something having high bias, but not really suscpible to training set (low variance) $\Rightarrow$ **Underfitting**.



If instead we use a model too complex we get something with very low bias, but extrimelly sensitive to the data we sample $\Rightarrow$ **Overfitting**.



What we see in practise is that Bias and Variance are strictly correlated: while one of the two decreases, the other increases. This force us to find some trade-off between these twos.